

A strict dg Lie proof of the Grünbaum–Sreedharan Stanley–Reisner smoothing theorem

Abstract

We prove that the Stanley–Reisner threefold associated with the Grünbaum–Sreedharan eight-vertex triangulated three-sphere is the special fibre of a flat projective scheme over $\mathbf{Q}[[s]]$ whose geometric generic fibre is smooth. The deformation controller is the strict dg Lie algebra of internal-weight-zero derivations of a Tate resolution relative to the fixed polynomial ring. An exact deformation over $\mathbf{Q}[s]/(s^4)$ is lifted directly to a square-zero Tate differential. The ordinary Maurer–Cartan equation, the strict Bianchi identity, and a certified exact sequence in degrees 1, 2, 3 then extend its prescribed two-jet to all orders. No transferred structure and no higher multilinear operation is used.

Contents

1	Statement and exact input	2
2	The strict controller and its classical meaning	2
2.1	Gradings and derivations	2
2.2	Maurer–Cartan elements give embedded flat algebras	3
3	The exact R_4 deformation is an actual Maurer–Cartan element	5
3.1	The certified presentation and flatness	5
3.2	A differential on the audited Tate resolution	6
4	The certified degree-1, 2, 3 exact sequence	7
4.1	What the 136 cycles certify	7
4.2	Why the computed columns are genuine H^3 classes	7
4.3	Exactness, with both maps computed	8
5	Strict Bianchi induction to all orders	9
6	The formal embedded family and its algebraization	10
7	The 280 incidence certificates and smoothness	11
8	Logical and computational dependency ledger	12

1 Statement and exact input

Put $k = \mathbf{Q}$, $S = k[x_1, \dots, x_8]$, and let Δ have facets

$$\begin{aligned} &1234, 1237, 1267, 1347, 1567, 2345, 2367, 3467, 3456, 4567, \\ &2358, 2368, 3568, 1268, 1568, 1248, 2458, 1478, 1578, 4578. \end{aligned}$$

Its Stanley–Reisner ideal is

$$\begin{aligned} I = (&x_6x_7x_8, x_4x_6x_8, x_3x_7x_8, x_3x_5x_7, x_3x_4x_8, x_2x_7x_8, \\ &x_2x_5x_7, x_2x_5x_6, x_2x_4x_7, x_2x_4x_6, x_1x_4x_6, x_1x_4x_5, \\ &x_1x_3x_8, x_1x_3x_6, x_1x_3x_5, x_1x_2x_5) \subset S. \end{aligned} \tag{1.1}$$

Write these ordered monomials as $m_1, \dots, m_{16}$, set $A = S/I$, and put $X_0 = \text{Proj } A \subset \mathbf{P}_k^7$. For $n \geq 1$ let

$$R_n = k[s]/(s^n), \quad R = k[[s]].$$

Theorem 1.1 (smoothing theorem). *There is a flat projective morphism*

$$X \longrightarrow \text{Spec } \mathbf{Q}[[s]]$$

whose special fibre is X_0 and whose geometric generic fibre is smooth. Consequently X_0 is smoothable in characteristic zero.

This is the exact conclusion proved here. In particular, no assertion about the canonical bundle, a Calabi–Yau structure, irreducibility, or a named smooth threefold is part of the theorem.

Verified computation 1.2 (special fibre). The exact rational computation `../referee-packet/code/verify_combinatorics.py` verifies that the displayed complex has 20 tetrahedral facets and precisely the sixteen minimal nonfaces in (1.1). It verifies Reisner’s vanishing condition on all 97 faces by exact boundary-matrix row reduction over $\mathbf{Q}$. Thus A is Cohen–Macaulay of Krull dimension 4, and X_0 is an equidimensional Cohen–Macaulay projective threefold. The deterministic output is `../referee-packet/verification/combinatorics_QQ.txt`.

2 The strict controller and its classical meaning

2.1 Gradings and derivations

Choose a full internally graded Tate resolution $P \rightarrow A$ over S which continues the independently certified low Tate stages described in Section 4. It is a semi-free graded-commutative dg S -algebra in nonnegative homological degrees, its differential d lowers homological degree by one and preserves internal weight, and

$$H_0(P) = A, \quad H_i(P) = 0 \quad (i > 0).$$

Tate resolutions exist by Avramov [1, Proposition 6.1.4]; over a $\mathbf{Q}$ -algebra the usual graded-commutative description agrees with the divided-power description needed in general.

A cohomological degree- p derivation lowers Tate homological degree by p . Define two strict dg Lie algebras, always retaining internal weight zero,

$$\mathfrak{g} = \text{Der}_k(P, P)_0, \quad \mathfrak{h} = \text{Der}_S(P, P)_0.$$

For $\theta \in \mathfrak{g}^p$ and $\eta \in \mathfrak{g}^q$, set

$$\partial\theta = [d, \theta] = d\theta - (-1)^p\theta d, \quad [\theta, \eta] = \theta\eta - (-1)^{pq}\eta\theta. \quad (2.1)$$

The commutator of derivations is again a derivation, and (2.1) gives strict dg Lie algebras.

The relative algebra $\mathfrak{h}$ is the correct fixed-coordinate controller: its degree-zero automorphisms fix S , hence fix the eight projective coordinates. The absolute algebra $\mathfrak{g}$ is nevertheless the one whose H^1 is the computed abstract tangent space. These statements are compatible, rather than competing, because

$$\mathfrak{h}^p = \mathfrak{g}^p \quad (p \geq 1). \quad (2.2)$$

Indeed, a positive-degree derivation sends $P_0 = S$ to a negative homological degree, which is zero. It follows that

$$H^1(\mathfrak{h}) \rightarrow H^1(\mathfrak{g}), \quad H^q(\mathfrak{h}) \xrightarrow{\sim} H^q(\mathfrak{g}) \quad (q \geq 2). \quad (2.3)$$

The first arrow need not be an isomorphism. In fact, the exact package calculation gives dimensions 109 and 53 for the relative embedded tangent space and the absolute tangent space respectively. We never identify them.

The augmentation induces a quasi-isomorphism

$$\mathrm{Der}_k(P, P)_0 \longrightarrow \mathrm{Der}_k(P, A)_0. \quad (2.4)$$

To see this directly, write the source and target as $\mathrm{Hom}_P(\Omega_{P/k}, P)_0$ and $\mathrm{Hom}_P(\Omega_{P/k}, A)_0$. The module of differentials of a semi-free algebra is semi-free, so Hom from it preserves the quasi-isomorphism $P \rightarrow A$; compare Avramov [1, Corollary 6.2.4]. Hence

$$H^q(\mathfrak{g}) = T_0^q(A). \quad (2.5)$$

The identical argument with $\Omega_{P/S}$ gives

$$\mathrm{Der}_S(P, P)_0 \longrightarrow \mathrm{Der}_S(P, A)_0 \quad \text{as a quasi-isomorphism.} \quad (2.6)$$

Here $\mathrm{Der}_S^0(P, A) = 0$, and a relative degree-one cocycle is exactly an S -linear map $I \rightarrow A$ of internal degree zero. Consequently $H^1(\mathfrak{h}) = \mathrm{Hom}_S(I, A)_0$. The exact computation in `verify_relative_controller.m2` obtains dimensions 109 for this space, 56 for the graded Jacobian/coordinate-boundary image, and 53 for the absolute quotient. The bracket used below is the bracket induced by the literal commutator in (2.1), not a product on a conormal Ext group.

2.2 Maurer–Cartan elements give embedded flat algebras

For R_n , let

$$\widehat{\mathfrak{h}}_n = \mathfrak{h} \otimes_k (s)/(s^n),$$

and over R let

$$\widehat{\mathfrak{h}} = s\mathfrak{h}[[s]], \quad F^r\widehat{\mathfrak{h}} = s^r\mathfrak{h}[[s]]. \quad (2.7)$$

This filtration is complete and separated, the differential preserves it, and brackets add filtration orders.

Lemma 2.1 (filtered lifting of a Tate differential). *Let $P_{<q}$ be the subalgebra generated by the Tate generators of homological degree less than q , and assume $H_{q-2}(P_{<q}, d) = 0$, where $q \geq 3$. Let D be a square-zero weight-preserving differential on $P_{<q} \otimes R_n$ with $D \equiv d \pmod{s}$. Every homogeneous d -cycle $c_0 \in (P_{<q})_{q-1}$ has a homogeneous lift $c \equiv c_0 \pmod{s}$ satisfying $Dc = 0$.*

Proof. Construct $c^{(r)} = c_0 + su_1 + \cdots + s^{r-1}u_{r-1}$ so that $Dc^{(r)} \in s^r P_{<q} \otimes R_n$. This holds for $r = 1$ because $dc_0 = 0$. Write

$$Dc^{(r)} = s^r \rho_r \pmod{s^{r+1}}.$$

The identity $D^2 = 0$ on $P_{<q}$ gives

$$0 = D(Dc^{(r)}) = s^r d\rho_r \pmod{s^{r+1}},$$

so $d\rho_r = 0$. The class of ρ_r lies in homological degree $q - 2$; by hypothesis choose a homogeneous u_r with $du_r = -\rho_r$. Then

$$c^{(r+1)} = c^{(r)} + s^r u_r$$

satisfies $Dc^{(r+1)} \in s^{r+1} P_{<q} \otimes R_n$. After the $(n-1)$ st step its image c in R_n is a D -cycle. Taking the internal homogeneous component matching c_0 at every step is legitimate because d and D preserve weight. $\square$

Proposition 2.2 (direct classical realization). *Let $\alpha \in \widehat{\mathfrak{h}}_n^1$ satisfy*

$$\partial\alpha + \frac{1}{2}[\alpha, \alpha] = 0. \tag{2.8}$$

Then $D = d + \alpha$ is a square-zero, $S \otimes R_n$ -linear, weight-preserving derivation on $P \otimes R_n$. If

$$B_n = H_0(P \otimes R_n, D),$$

then B_n is a flat graded embedded deformation of A in the fixed coordinates. If $e_1, \dots, e_{16}$ are the degree-one Tate generators with $d(e_i) = m_i$, then, literally,

$$B_n = R_n[x_1, \dots, x_8] / (D(e_1), \dots, D(e_{16})), \tag{2.9}$$

and every $D(e_i)$ is a cubic.

Conversely, every marked flat graded quotient of $R_n[x_1, \dots, x_8]$ specializing to A admits, after choices of homogeneous Tate lifts, a differential $D = d + \alpha$ of this form. Conjugating D by a nilpotent relative degree-zero automorphism changes the semi-free trivialization while fixing S and induces an isomorphic embedded deformation. No converse classification of all isomorphisms by gauge is needed below.

Proof. For a degree-one derivation,

$$(d + \alpha)^2 = \partial\alpha + \frac{1}{2}[\alpha, \alpha],$$

so (2.8) is exactly $D^2 = 0$. Relative derivations vanish on S , and weight zero makes $D(e_i)$ homogeneous of internal degree three. Since every homological-degree-one element is an $S \otimes R_n$ -linear combination of the e_i , the degree-zero boundaries form exactly the ideal in (2.9).

Filter $(P \otimes R_n, D)$ by powers of s . The filtration is finite, and its associated-graded differential is d . Its spectral sequence has

$$E_1 = H(P) \otimes_k \text{gr}(R_n) = A \otimes_k \text{gr}(R_n)$$

in homological degree zero and is zero in positive homological degrees. Thus $(P \otimes R_n, D)$ is exact in positive degrees. In each internal degree it is a free R_n -resolution of the finite module $(B_n)_j$. Tensoring this resolution with k gives P_j , so

$$\text{Tor}_1^{R_n}(k, (B_n)_j) = H_1(P_j) = 0.$$

The local criterion for flatness [4, Tag 00MK] makes every $(B_n)_j$ flat, hence free, over the Artinian local ring R_n .

For the converse, let J be the kernel of a marked flat quotient $R_n[x] \twoheadrightarrow B_n$. Degreewise base change in

$$0 \longrightarrow J \longrightarrow R_n[x] \longrightarrow B_n \longrightarrow 0$$

is exact. Lift the basis $m_1, \dots, m_{16}$ of I_3 to elements $F_i \in J_3$. Nakayama's lemma, first in degree three and then in every degree, says that the F_i generate J . The kernel of

$$R_n[x](-3)^{16} \xrightarrow{(F_1, \dots, F_{16})} R_n[x]$$

also commutes with reduction, as follows in two steps. In $0 \rightarrow J \rightarrow R_n[x] \rightarrow B_n \rightarrow 0$, both the middle and right modules are R_n -flat, so J is R_n -flat. If K is the displayed first-syzygy kernel, then $0 \rightarrow K \rightarrow R_n[x](-3)^{16} \rightarrow J \rightarrow 0$ and $\mathrm{Tor}_1^{R_n}(k, J) = 0$ identify $K \otimes k$ with the full special kernel. Hence the thirty special first syzygies lift to genuine relations among the F_i . Define D on e_i and f_j by these equations and lifted relations.

It remains only to extend D over the later Tate generators. If z has homological degree $q \geq 3$, then dz is a cycle of degree $q - 1$ in $P_{<q}$. By the staged Tate construction, all homology through degree $q - 2$ has already been killed. Lemma 2.1 therefore gives a D -cycle $c_z \equiv dz \pmod{s}$; set $D(z) = c_z$. Induction in the Tate generators constructs D . Since the square of an odd derivation is again a derivation, checking D^2 on the generators proves $D^2 = 0$ everywhere. The difference $D - d$ is the required relative Maurer–Cartan element. A nilpotent exponential of a relative degree-zero derivation conjugates D and gives the stated one-way gauge assertion. $\square$

The converse is used below in an even stronger form: the relevant R_4 differential is constructed explicitly through the first relations, and the remaining extension is the last paragraph of the proof. Thus no abstract comparison of deformation hulls is invoked.

3 The exact R_4 deformation is an actual Maurer–Cartan element

3.1 The certified presentation and flatness

The exact tangent direction in the ordered 53-element absolute tangent basis is

$$\begin{aligned} y = (& 1, 3, 1, 2, 6, 1, 1, 1, 1, 1, 21, 28, 1, 5, 10, 55, 66, 15, 1, 18, 2, \\ & 35, 42, 1, 4, 10, 1, 25, 12, 30, 5, 33, 44, 1, 6, 3, 14, 7, 1, 9, 1, 6, \\ & 12, 20, 1, 4, 22, 1, 15, 1, 1, 8, 11). \end{aligned} \tag{3.1}$$

The pinned calculation produces matrices F_0, F_1, F_2, F_3 and Q_0, Q_1, Q_2, Q_3 . After substituting $t = sy$, set

$$F^{[3]} = \sum_{i=0}^3 F_i, \quad Q^{[3]} = \sum_{i=0}^3 Q_i$$

over $R_4[x_1, \dots, x_8]$.

Verified computation 3.1 (starting deformation). The script `../completion/verify_starting_jet.m2` verifies, by exact arithmetic over $\mathbf{Q}$, that

$$F^{[3]}Q^{[3]} = 0, \tag{3.2}$$

that Q_0 is the complete thirty-column first-syzygy matrix of F_0 , and that $F^{[3]} \bmod s^3$ is byte-for-byte the sixteen-row file `../referee-packet/data/universal_2jet_QQ.txt`. Its SHA-256 is

$$40e64e61674b6a4e61f1ea6822dc79327bf4ba397285f84ed8738c4c18cd1795. \quad (3.3)$$

The certificate is `../completion/verification/starting_jet_QQ.txt`.

Let

$$A_4 = R_4[x]/(F^{[3]}).$$

This is an actual flat algebra, not merely a matrix satisfying a formal equation. In an internal degree j , let K_j be the kernel of the map defined by $F^{[3]}$. Since the columns of $Q^{[3]}$ lie in K_j and reduce to a generating set of the entire special kernel, the reduction map

$$K_j \otimes_{R_4} k \longrightarrow \ker(F_0)_j$$

is onto. Extend the presentation on the left by a free module. After tensoring with k , this surjectivity says precisely that $\mathrm{Tor}_1^{R_4}(k, (A_4)_j) = 0$. The local criterion [4, Tag 00MK] makes $(A_4)_j$ free. This holds in every internal degree, so A_4 is a flat graded embedded deformation. Its quotient $A_4/s^3 A_4$ is the printed two-jet.

3.2 A differential on the audited Tate resolution

We now construct the promised Maurer–Cartan representative rather than deducing one from the package terminology. The certified beginning of P has generators

	homological degree	internal weight
$e_1, \dots, e_{16}$	1	3
$f_1, \dots, f_{30}$	2	4
$g_1, \dots, g_{136}$	3	5 (16 times), 6 (120 times).

(3.4)

Let R_0 be the relation matrix used for $d(f)$ in this Tate model. The strict bracket script computes an invertible constant matrix U with

$$Q_0 U = R_0, \quad \det U = 1. \quad (3.5)$$

On $P \otimes R_4$, define

$$D(e_i) = F_i^{[3]}, \quad D(f_j) = \sum_i (Q^{[3]} U)_{ij} e_i. \quad (3.6)$$

Equations (3.2) and (3.5) give $D^2(e_i) = D^2(f_j) = 0$, and $D \equiv d \pmod{s}$.

Extend D over each g_a and then every later Tate generator by Lemma 2.1. For a generator of homological degree $q \geq 3$, the obstruction at each s -order is a d -cycle in degree $q - 2 > 0$ and has a homogeneous primitive in degree $q - 1$, which cannot involve that generator or a later one. The completeness of the thirty f -stage and the 136-generator g -stage needed here is independently certified in Section 4; all later stages are Tate stages by construction. We obtain a square-zero differential on the same underlying semi-free algebra,

$$D = d + \bar{\alpha}, \quad \bar{\alpha} \in \mathrm{MC}(\mathfrak{h} \otimes (s)/(s^4)). \quad (3.7)$$

Its degree-zero homology is exactly A_4 , because its degree-one boundaries are the sixteen entries of $F^{[3]}$. Its linear coefficient is the strict degree-one cocycle whose values on the e_i are $T^1 y$; its image in $H^1(\mathfrak{g})$ is therefore the computed class y . This proves the required connection between the R_4 matrices and an actual truncated Maurer–Cartan element.

4 The certified degree-1, 2, 3 exact sequence

4.1 What the 136 cycles certify

Let D_1, D_2 be the chain maps in homological degrees one and two of the materialized Tate algebra. Let D_3 denote the differentials of the decomposable degree-three elements $e_i f_j$ and $e_i e_j e_k$, and let Z be the 150×136 matrix whose columns are the proposed $d(g_a)$.

Verified computation 4.1 (complete low Tate stages). The independent audit in `../audits/tate-stage/` reconstructs all four matrices from the canonical sparse input and proves over $\mathbf{Q}[x_1, \dots, x_8]$

$$D_1 D_2 = 0, \quad D_2 D_3 = 0, \quad D_2 Z = 0,$$

and the unbounded graded-module equalities

$$\ker D_1 = \operatorname{im} D_2, \quad \ker D_2 = \operatorname{im}[D_3 \ Z]. \quad (4.1)$$

Macaulay2 computes full homogeneous syzygy modules and two-sided lifts; Singular independently recomputes the modules and lifts. The result is `../audits/tate-stage/MATHEMATICAL_CERTIFICATE.md`.

The precise meaning of (4.1) is worth emphasizing. The 136 columns are cycles in homological degree two. Their classes span

$$\ker D_2 / \operatorname{im} D_3, \quad (4.2)$$

so adjoining the g_a kills all remaining degree-two homology. They are not asserted to be a basis of H^3 , and no total dimension of H^3 is used. Equation (4.1) proves that no additional Tate generators of homological degree at most three are missing; a full Tate resolution can continue with generators of degree at least four.

4.2 Why the computed columns are genuine H^3 classes

Let

$$C^p = \operatorname{Der}_k^p(P, A)_0.$$

On (3.4),

$$C^1 = (A_3)^{16}, \quad C^2 = (A_4)^{30}, \quad C^3 = (A_5)^{16} \oplus (A_6)^{120}. \quad (4.3)$$

The exact bracket calculation produces a closed degree-one derivation θ representing y and closed degree-two derivations $\eta_1, \dots, \eta_{27}$ representing a basis of H^2 . It solves and rechecks their closure equations on every e_i, f_j, g_a . It also constructs degree-one derivations $\psi_1, \dots, \psi_{53}$ for the absolute tangent basis and checks their closure through every f_j ; their later extension is included in the next lemma.

Lemma 4.2 (extension of the computed derivations). *Each of $\theta, \eta_1, \dots, \eta_{27}$ extends, without changing its values on (3.4), to a closed derivation of the full Tate resolution. Each ψ_i extends without changing its values on the e - and f -generators.*

Proof. For a ψ_i , the first unchecked generators are the g_a of degree three. The right side of its closure equation is a degree-one cycle. The equality $\ker D_1 = \operatorname{im} D_2$ in Computation 4.1 says exactly that this cycle has a degree-two primitive in the algebra generated by the e and f . Thus every ψ_i extends through all the g_a without changing the values used by the primary-commutator calculation.

Suppose a degree- p derivation ϑ , with $p = 1$ or 2 , is already closed on earlier generators, and let z be a new Tate generator of degree $q \geq 4$. Closedness on z requires

$$d(\vartheta z) = (-1)^p \vartheta(dz). \quad (4.4)$$

The right side is a cycle of homological degree $q - p - 1 > 0$. It is a boundary because P is acyclic in positive degree. A primitive has degree $q - p < q$, hence uses only earlier generators. Its component of the required internal weight is still a primitive. Induction proves the assertion. The low cases excluded by the inequality are exactly the closure equations checked on (3.4). $\square$

The commutators $[\theta, \eta_j]$ are therefore global closed degree-three derivations. Under (2.4), their cochains lie in the kernel of the outgoing map $C^3 \rightarrow C^4$. Consequently

$$H^3(C) = \ker(C^3 \rightarrow C^4) / \text{im}(C^2 \rightarrow C^3) \hookrightarrow C^3 / \text{im}(C^2 \rightarrow C^3). \quad (4.5)$$

Independence of these already-closed columns in the quotient on the right is therefore independence in genuine H^3 . This is the rigorous role of the 136-cycle certificate; a bare degree-three generator count would not prove (4.5).

4.3 Exactness, with both maps computed

Proposition 4.3 (fixed-direction strict exactness). *Let $a \in Z^1(\mathfrak{h})$ be the linear coefficient of (3.7). Then*

$$H^1(\mathfrak{h}) \xrightarrow{\text{ad}_a} H^2(\mathfrak{h}) \xrightarrow{\text{ad}_a} H^3(\mathfrak{h}) \quad (4.6)$$

is exact at H^2 . The image of the first map has dimension 15 and the second map has rank 12 on the 27-dimensional middle space.

Proof. The script `../referee-packet/code/verify_aq_bracket.m2` performs the following exact operations.

First relate its chosen cocycle θ to the coefficient a of the actual differential (3.7). Both have absolute cohomology class y , by the last paragraph of Section 3, so

$$a - \theta = \partial \xi \quad \text{for some } \xi \in \mathfrak{g}^0. \quad (4.7)$$

If z is any closed relative derivation of positive degree, then

$$[a - \theta, z] = [\partial \xi, z] = \partial[\xi, z]. \quad (4.8)$$

The cochain $[\xi, z]$ has positive degree and is therefore relative by (2.2). Thus ad_a and ad_θ induce the same maps on relative H^1 and H^2 . In particular, it is enough to compute with the literal θ used by the script; this is a chain-level comparison, not an appeal to invariance of any transferred operation.

First, it transports all 27 package obstruction columns through the constant relation-basis change (3.5), checks that they are closed, proves that they are independent modulo boundaries, and proves that no further internal-degree-zero H^2 cocycle remains. Thus they are a basis of the 27-dimensional H^2 .

Second, it computes the literal graded commutators

$$[\theta, \eta_j] = \theta \eta_j - \eta_j \theta$$

on all 136 generators g_a . By Lemma 4.2 and (4.5), the rank 12 found modulo every degree-three boundary is the rank of

$$A_y = \text{ad}_y : H^2(\mathfrak{g}) \longrightarrow H^3(\mathfrak{g}).$$

Third, it separately computes the primary commutators of θ with all 53 degree-one absolute tangent basis derivations. For two degree-one derivations the bracket is the signed sum $\theta\psi + \psi\theta$. The resulting map

$$B_y = \text{ad}_y : H^1(\mathfrak{g}) \longrightarrow H^2(\mathfrak{g})$$

has rank 15. More precisely, the script proves the matrix identity

$$B_y = -J Dq_y, \tag{4.9}$$

where J is the verified isomorphism from the 27 package obstruction coordinates to the displayed H^2 basis. It also checks directly, on those strict commutator matrices,

$$A_y B_y = 0. \tag{4.10}$$

Thus (4.10), not the rank of Dq_y alone, gives $\text{im } B_y \subseteq \ker A_y$. Rank-nullity gives

$$\dim \ker A_y = 27 - 12 = 15 = \dim \text{im } B_y,$$

so equality follows.

Finally transfer this statement from the absolute algebra to the relative controller. By (2.2)–(2.3), H^2 and H^3 and the second map are literally the same, while every absolute degree-one class has a relative cocycle representative. Conversely, the extra relative classes become absolute boundaries. If such a class is represented by $\partial\xi$, with $\xi \in \mathfrak{g}^0$, then

$$[a, \partial\xi] = -\partial[a, \xi], \tag{4.11}$$

because $\partial a = 0$; moreover $[a, \xi]$ has positive degree and is therefore relative. Hence the extra classes contribute zero to H^2 , and the images of the relative and absolute first maps coincide. Exactness of (4.6) follows. $\square$

The deterministic output `../referee-packet/verification/aq_bracket_QQ.txt` records the 27 classes, ranks 12 and 15, the zero composite, and the primary identity (4.9). Notice that the proof neither asserts nor needs a dimension for the whole group H^3 .

5 Strict Bianchi induction to all orders

For $\gamma \in s\mathfrak{h}^1[[s]]$, define its curvature by

$$\mathcal{F}(\gamma) = \partial\gamma + \frac{1}{2}[\gamma, \gamma]. \tag{5.1}$$

The dg Lie axioms give the strict Bianchi identity

$$\partial\mathcal{F}(\gamma) + [\gamma, \mathcal{F}(\gamma)] = 0. \tag{5.2}$$

Indeed, $\partial(\frac{1}{2}[\gamma, \gamma]) = -[\gamma, \partial\gamma]$ and graded Jacobi gives $[\gamma, [\gamma, \gamma]] = 0$.

Lemma 5.1 (fixed-two-jet extension). *Let $\bar{\alpha} \in \text{MC}(\mathfrak{h} \otimes (s)/(s^4))$ have linear coefficient a . If (4.6) is exact at H^2 , then there is*

$$\alpha_\infty \in \text{MC}(s\mathfrak{h}[[s]]) \quad \text{with} \quad \alpha_\infty \equiv \bar{\alpha} \pmod{s^3}. \tag{5.3}$$

Proof. Lift $\bar{\alpha}$ arbitrarily to $\gamma = sa + O(s^2)$. Its curvature is $O(s^4)$. Suppose, for $N \geq 4$, that

$$\mathcal{F}(\gamma) = s^N r_N + s^{N+1} r_{N+1} + O(s^{N+2}). \quad (5.4)$$

The coefficients of s^N and s^{N+1} in (5.2) give

$$\partial r_N = 0, \quad \partial r_{N+1} + [a, r_N] = 0. \quad (5.5)$$

Thus the obstruction coefficient is a cocycle and its class lies in the kernel of $\text{ad}_a : H^2 \rightarrow H^3$. Exactness supplies a closed $v \in \mathfrak{h}^1$ such that

$$[r_N] + [a, v] = 0.$$

Choose $w \in \mathfrak{h}^1$ with

$$r_N + [a, v] = \partial w. \quad (5.6)$$

Replace

$$\gamma \quad \text{by} \quad \gamma' = \gamma + s^{N-1}v - s^N w. \quad (5.7)$$

The exact change formula

$$\mathcal{F}(\gamma') - \mathcal{F}(\gamma) = \partial(\gamma' - \gamma) + [\gamma, \gamma' - \gamma] + \frac{1}{2}[\gamma' - \gamma, \gamma' - \gamma]$$

shows that the order- s^N change is $[a, v] - \partial w$, which cancels r_N by (5.6). All other terms have order at least $N + 1$: $\partial v = 0$, $\gamma - sa = O(s^2)$, and $2N - 2 \geq N + 1$.

The corrections are Cauchy in the filtration (2.7), so they converge coefficientwise. Continuity of (5.1), together with $\bigcap_N s^N \mathfrak{h}[[s]] = 0$, gives zero curvature in the limit. At the initial $N = 4$ step the correction is $s^3 v - s^4 w$; consequently the coefficients of s and s^2 are unchanged, proving (5.3). $\square$

Apply the lemma to (3.7) and Proposition 4.3. This gives α_∞ . It is important that the conclusion preserves the R_3 reduction, not necessarily the whole R_4 element: the first correction may change the coefficient of s^3 . Preserving the printed first and second-order terms is exactly what the smoothness certificate requires.

6 The formal embedded family and its algebraization

Put $D_\infty = d + \alpha_\infty$ and define the literal cubics

$$F_i(s) = D_\infty(e_i) \in R[x_1, \dots, x_8]_3. \quad (6.1)$$

By (5.3) and Computation 3.1,

$$F_i(s) = F_i^{(2)} + s^3 H_i(s, x), \quad (6.2)$$

where the sixteen $F_i^{(2)}$ are exactly the rows of the frozen two-jet and H_i is homogeneous cubic in x . There are only finitely many cubic monomials, so (6.1) is an honest element of $R[x]$, not a second completion in the x -variables.

Let

$$B = R[x_1, \dots, x_8]/(F_1(s), \dots, F_{16}(s)), \quad X = \text{Proj } B. \quad (6.3)$$

For every n , Proposition 2.2 applied to $\alpha_\infty \bmod s^n$ gives

$$B/s^n B = H_0(P \otimes R_n, D_\infty \bmod s^n), \quad (6.4)$$

and every internal-degree piece of (6.4) is free over R_n .

We check flatness over R , rather than assuming it from formal effectivity. Each B_j is a finite R -module. If $sb = 0$ in B_j , then flatness of $B_j/s^n B_j$ over R_n implies

$$b \in s^{n-1} B_j \quad \text{for every } n \geq 2,$$

because the annihilator of s on a free R_n -module is s^{n-1} times that module. Krull intersection gives $b = 0$. Hence B_j is torsion-free, and therefore free over the DVR R . The direct sum $B = \bigoplus_j B_j$ is R -flat. On a standard projective chart, $(B_{x_i})_0$ is a direct summand of the flat localization B_{x_i} , so $X \rightarrow \operatorname{Spec} R$ is flat.

Equation (6.3) is also an explicit algebraization of the compatible formal family (6.4): it is a closed subscheme of $\mathbf{P}_R^7$, is projective over R , and has exactly those reductions. Alternatively, the compatible closed subschemes satisfy every hypothesis of the Stacks Project algebraization lemma [5, Tag 0899]: R is Noetherian and (s) -adically complete, $\mathbf{P}_R^7$ is separated of finite type, the squares are cartesian, and X_0 is proper. The explicit cubics identify that algebraized closed subscheme with (6.3). Thus no formal family is being silently substituted for an algebraic one.

7 The 280 incidence certificates and smoothness

On the projective chart $x_v = 1$, write $z_1, \dots, z_7$ for the other affine coordinates and let J be the 7×16 affine Jacobian of the cubics. A four-plane in $\ker J^\top$ is represented on one of the 35 standard charts of $\operatorname{Gr}(4, 7)$ by a 7×4 matrix K_P whose pivot rows form the identity. For each of the $8 \cdot 35 = 280$ pairs, the truncated singular-incidence ideal has the 80 displayed generators

$$\mathcal{J}_{v,P}^{(2)} = (F_1^{(2)}, \dots, F_{16}^{(2)}, (J^{(2)})^\top K_P). \quad (7.1)$$

Verified computation 7.1 (all incidence charts). The scripts `../referee-packet/code/verify_incidence_chart.m2` and `../referee-packet/code/verify_all_incidence.py` prove, over exact $\mathbf{Q}$ arithmetic, that on every chart

$$s^2 \in (\mathcal{J}_{v,P}^{(2)}, s^3). \quad (7.2)$$

They encode the 80 generators of (7.1) over $\mathbf{Q}[z_1, \dots, z_7, a_1, \dots, a_{12}][s]/(s^3)$ as a 3×240 module matrix and prove that $(0, 0, 1)^\top$ belongs to its column module. Exact partial Gröbner reduction proves 277 charts at degree limit six; degree limit eight proves precisely $(v, P) = (2, 7), (6, 12)$, and $(8, 16)$. The driver verifies that all 280 pairs occur and checks one retained lift back to the original columns. The certificate is `../referee-packet/verification/incidence_all_QQ.txt`.

The certificates assert exactly (7.2). They do not, by themselves, assert that a fibre is smooth; that conclusion is the following geometric argument.

Proposition 7.2 (two-jet smoothness). *Every flat projective continuation satisfying (6.2) has smooth geometric generic fibre.*

Proof. The special fibre is Cohen–Macaulay and equidimensional of dimension three by Computation 1.2. For a proper flat family over R , the Cohen–Macaulay fibre criterion and constancy of relative dimension make the morphism Cohen–Macaulay of relative dimension three; these are exactly the opening reductions in the Stacks smoothing lemma [6, Tag 0E7T]. Thus a point of a fibre is singular precisely when the affine Jacobian has rank at most three, equivalently when $\ker J^\top$ contains a four-plane.

Suppose the geometric generic fibre has a singular point. The incidence scheme is of finite type over $K = \mathbf{Q}((s))$, so it has a closed point after a finite extension K'/K . Let R' be a DVR of the integral closure of R in K' lying over (s) . Properness of $\mathbf{P}_R^7$ extends the point over R' , and properness of $\mathrm{Gr}(4, 7)_R$ extends a chosen four-plane in the kernel. After scaling, some projective coordinate and some Plücker coordinate are units. Hence one of the 280 chart pairs has all of its point and four-plane coordinates in R' .

On this chart, (7.2) is an exact polynomial identity

$$s^2 = \sum_j c_j q_j + s^3 c_0, \quad (7.3)$$

where the q_j are the truncated incidence generators. Equation (6.2) changes each equation, and each affine partial derivative, only by s^3 times an integral power series. Replacing q_j by the actual incidence generator q'_j in (7.3) therefore gives

$$s^2 = \sum_j c_j q'_j + s^3 c$$

with $c \in R'$ after evaluation at the integral incidence section. All q'_j vanish there, so $s^2 = s^3 c$. The valuation of s in R' is positive, whereas this equality would give

$$2v(s) = 3v(s) + v(c) \geq 3v(s),$$

a contradiction. The geometric generic fibre is smooth. $\square$

Proof of Theorem 1.1. Computation 3.1 and the direct Tate lift (3.6)–(3.7) give a truncated relative Maurer–Cartan element with the certified two-jet. Proposition 4.3 and Lemma 5.1 produce the all-orders strict Maurer–Cartan element without changing that two-jet. Equations (6.1)–(6.4) give a flat projective family with special fibre X_0 , and Proposition 7.2 proves that its geometric generic fibre is smooth. $\square$

8 Logical and computational dependency ledger

Statement	Evidence	Exact consequence used
Special fibre	Exact combinatorial replay; Reiser’s criterion	Pure CM threefold with ideal (1.1)
R_4 start	Exact $F^{[3]}Q^{[3]} = 0$, complete special syzygies, byte comparison	Flat embedded deformation and literal two-jet
Low Tate stage	Independent Macaulay2 and Singular full syzygy modules	136 cycles span (4.2); no missing low generators
Strict exactness	Closed global derivations; all primary and secondary commutators; ranks and direct zero composite	Exactness (4.6), without a total H^3 dimension
All orders	Equations (5.1)–(5.7)	Complete relative Maurer–Cartan element fixing coefficients of s, s^2
Generic smoothness	280 exact memberships plus projective/Grassmannian properness	Valuation contradiction for every higher continuation
Algebraization	Sixteen coefficientwise cubic series; alternatively Tag 0899	Actual projective R -scheme, not only formal data

The computations do not prove a total value for $\dim H^3$, do not preserve the full s^3 coefficient of the starting R_4 object, and do not identify the generic fibre beyond its smoothness. None of those stronger statements is needed.

References

- [1] L. L. Avramov, *Infinite free resolutions*, in *Six Lectures on Commutative Algebra*, Progress in Mathematics 166, Birkhäuser, 1998, pp. 1–118.
- [2] B. Grünbaum and V. P. Sreedharan, *An enumeration of simplicial 4-polytopes with 8 vertices*, J. Combinatorial Theory **2** (1967), 437–465, doi:10.1016/S0021-9800(67)80055-3.
- [3] G. A. Reisner, *Cohen–Macaulay quotients of polynomial rings*, Adv. Math. **21** (1976), 30–49, doi:10.1016/0001-8708(76)90114-6.
- [4] The Stacks Project Authors, *The Stacks Project*, Tag 00MK, local criterion for flatness.
- [5] The Stacks Project Authors, *The Stacks Project*, Tag 0899, Grothendieck algebraization for compatible closed subschemes.
- [6] The Stacks Project Authors, *The Stacks Project*, Tag 0E7T, smoothing lemma.